FINAL-PROJECT

WANG TIANGE 3036766442

1. Introduction

1.1 Research Question

This project aims to replicate and extend the findings of Card and Krueger (1994), "Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania." The central research question is: Does an increase in the minimum wage reduce employment in low-wage industries?

1.2 Introduce the Experiment

On April 1, 1992, New Jersey (NJ) raised its minimum wage from $4.25 to $5.05 per hour, giving it the highest state minimum wage in the country at that time. In contrast, the neighboring state of Pennsylvania (PA) kept its minimum wage constant at $4.25 per hour. Because these two states share similar economic conditions, seasonal patterns, and labor markets, the policy change in NJ provides an ideal setting to evaluate the employment effects of a minimum wage increase. By comparing fast-food restaurants in NJ and PA before and after the policy change, Card and Krueger were able to isolate the causal effect of the minimum wage from other confounding factors.

2. Sample Design and Evaluation

2.1 Data Sources

This project utilizes the original survey data from Card and Krueger (1994), which was collected to evaluate the impact of New Jersey's minimum wage increase. The dataset contains establishment-level data from 410 fast-food restaurants in New Jersey (NJ) and eastern Pennsylvania (PA). The data was collected through two telephone surveys: Wave 1 was conducted in February and March 1992 that was before the policy change, and Wave 2 was conducted in November and December 1992 , approximately 8 months after the policy change. The dataset includes information on employment, starting wages, product prices, store characteristics, and closure status.

2.2 Sample Construction and Screening

The raw dataset contains 410 observations. To construct the analysis sample, we implemented the following screening and cleaning procedures, strictly following the methodology described in the original paper:

Time Alignment: The data is already structured into two distinct waves (Wave 1 and Wave 2), corresponding to the pre- and post-policy periods. We matched the two waves by the unique store identifier (SHEET) to create a panel structure.

Handling Missing Values and Closures: For the employment analysis, we distinguished between permanently closed stores and temporarily closed stores. Based on the STATUS2 variable, we set the Wave 2 employment (FTE) to 0 for stores that were permanently closed (Code 3).

Finally, we remove observations with missing employment information and conduct our analysis using the remaining 365 observations.

2.3 Variable Definitions

The key variables used in the replication are defined as follows:

Sheet: Original store/survey sheet identifier.

Chain: 1 = Burger King; 2 = KFC; 3 = Roy Rogers; 4 = Wendy’s.

Company ownership indicator: 1 = company-owned.

State: 1 = New Jersey; 0 = Pennsylvania.

EMPFT: Number of full-time employees

EMPPT: Number of part-time employees

NMGRS: Number of managers and assistant managers

WAGE_ST: Starting wage, in dollars per hour.

HRSOPEN: Hours open per day.

3.Empirical Design

To estimate the causal effect of the minimum wage increase on employment, we employ a Difference-in-Differences (DID) framework. The empirical design exploits the natural experiment created by New Jersey's minimum wage increase in April 1992, while neighboring Pennsylvania's minimum wage remained constant at $4.25 per hour.

3.1 Cross-State Comparison (DID)

The first specification compares the change in employment at fast-food stores in New Jersey (the treatment group) to the change in employment at stores in Pennsylvania (the control group). The regression model is specified as:

$$\Delta E_i = \alpha + \beta NJ_i + \gamma X_i + \epsilon_i$$

Where:

E_i is the change in Full-Time Equivalent (FTE) employment at store $i$between Wave 1 and Wave 2.

NJ_i is a binary indicator equal to 1 if the store is located in New Jersey, and 0 if in Pennsylvania. The coefficient βcaptures the DID estimate of the minimum wage effect.

X_i is a vector of control variables, including chain dummies and a company-owned dummy.

ϵ_i is the error term.

3.2 Within-New Jersey Comparison (Wage Gap)

The second specification exploits variation in initial starting wages within New Jersey. We replace the New Jersey dummy with a continuous "Wage Gap" variable (GAP_i), which measures the proportional wage increase required for a store to reach the new minimum wage of $5.05. The model is:

$$\Delta E_i = \alpha' + \beta' GAP_i + \gamma' X_i + \epsilon_i'$$

Where:

$GAP_i = 0$ for stores in Pennsylvania and for New Jersey stores already paying $5.05 or more.

$GAP_i = (5.05 - WAGE_ST_i)/WAGE_ST_i$ for other New Jersey stores.

The coefficient β' measures the effect of the minimum wage on employment, identified by comparing high-wage and low-wage stores within New Jersey.

4. Core Replication Results and Comparison

4.1 Summary Statistics (Table 2)

Table 2 in the original essay presents the means of key variables for New Jersey and Pennsylvania stores in both waves.

Table 1 in my project presents the summary statistics of store characteristics and core variables for New Jersey (NJ, treatment group) and Pennsylvania (PA, control group) in Wave 1 (pre-policy)

and Wave 2 (post-policy).

In Wave 1 before the minimum wage reform, the composition of restaurant brands and the share of company-owned stores are broadly similar between the two states. NJ and PA have comparable starting wages, share of stores paying the old minimum wage ($4.25), full meal prices and weekday opening hours. The t-statistics for most variables are small, meaning there is **no statistically significant difference** between NJ and PA before the policy shock. This supports the parallel pre-trend assumption for the DID design.

After the policy implementation in Wave 2, the starting wage in New Jersey rises sharply to 5.08, and no NJ stores remain at the old $4.25 wage floor. Meanwhile, Pennsylvania's starting wage stays nearly unchanged. The FTE employment in NJ rises slightly, while PA employment falls, consistent with the baseline DID findings.

Table 1 Means of Key Variables

Variable	NJ	PA	t^2
1.Distribution of Store Types (percentages):			
Burger King	41.09	44.30	
KFC	20.54	15.19	
Roy Rogers	24.77	21.52	
Wendys	13.60	18.99	
Company-owned	34.14	35.44	
2. *Means in Wave I:*			
FTE employment	20.44 (0.51)	23.33 (1.35)	2
Percentage full-time employees	32.85 (1.33)	35.04 (2.73)	1
Starting wage	4.61 (0.02)	4.63 (0.04)	0
Wage = 4.25 (percentage)	30.51 (2.53)	32.91 (5.32)	0
Price of full meal	3.35 (0.04)	3.04 (0.07)	-4
Hours open (weekday)	14.42 (0.15)	14.53 (0.33)	0
Recruiting bonus	23.56 (2.34)	29.11 (5.14)	1

3. *Means in Ware 2:*			
	21.03	21.17	
FTE employment	(0.52)	(0.94)	0
	35.87	30.38	
Percentage full-time employees	(1.38)	(2.81)	-2
	5.08	4.62	
Starting wage	(0.01)	(0.04)	-20
		25.32	
Wage = 4.25 (percentage)	0.00 (0.00)	(4.92)	11
		0.00	
Wage = 5.05 (percentage)	0.00 (0.00)	(0.00)	
	3.41	3.03	
Price of full meal	(0.04)	(0.07)	-5

Notes: See text for definitions. Standard errors are given in parentheses.

Test of equality of means in New Jersey and Pennsylvania.

4.2 Employment Effects (Table 3)

Table 3 reports the average change in FTE employment by state and by initial wage group within New Jersey. Table 3 of origin reports the average change in FTE employment by state and by initial wage group within New Jersey.

Table 2 of mine represents the average FTE employment per store before and after the minimum wage increase, and the changes in employment for Pennsylvania (PA), New Jersey (NJ), and subgroups within New Jersey classified by pre-policy wage levels.

Row 1 and Row 2 show the pre- and post-policy FTE employment for the full unbalanced sample. Before the policy, PA stores had higher average FTE employment (23.33) than NJ stores (20.44). After the minimum wage hike, PA employment fell to 21.17 while NJ employment rose slightly to 21.03. Row 3 reports the simple DID estimate based on all available observations: the relative employment growth of NJ compared with PA is 2.75 FTE.

Rows 4 and 5 repeat the calculation using balanced samples, yielding similar positive DID coefficients around 2.75 and 2.51, which confirms that the core finding is not driven by unbalanced panel attrition. Within New Jersey, stores with the lowest initial wages gained employment, whereas high-wage NJ stores experienced employment declines. This heterogeneous pattern supports the argument that the policy effect comes from the minimum wage shock rather than common regional trends.

Table 2 Average Employment per Store Before and After the Rise in New Jersey Minimum Wage

Variable	PA	NJ	Difference NJ-PA	Wage=$4.25	Wage=$4.26-$4.99	Wage>=$5.00	Low-high	Midrange-high
1. FTE employment before,all available observations	23.33 (1.35)	20.44 (0.51)	-2.89 (1.44)	19.56 (0.77)	20.08 (0.84)	21.99 (1.00)	-2.43 (1.26)	-1.91 (1.31)
2. FTE employment after, all available observations	21.17 (0.94)	21.03 (0.52)	-0.14 (1.08)	20.88 (1.01)	20.96 (0.76)	21.31 (1.01)	-0.43 (1.43)	-0.36 (1.26)
3. Change in mean FTE employment	-2.17 (1.65)	0.59 (0.73)	2.75 (1.80)	1.32 (1.27)	0.87 (1.13)	-0.68 (1.42)	2.00 (1.91)	1.55 (1.82)
4. Change in mean FTE employment, balanced sample of stores	-2.28 (1.25)	0.47 (0.48)	2.75 (1.34)	1.20 (0.82)	0.71 (0.69)	-0.76 (1.07)	1.96 (1.35)	1.47 (1.27)
5. Change in mean FTE employment, setting FTE at temporarily closed stores to 0	-2.28 (1.25)	0.23 (0.49)	2.51 (1.35)	0.90 (0.87)	0.49 (0.69)	-0.96 (1.08)	1.86 (1.38)	1.46 (1.28)

Notes: Standard errors are shown in parentheses. FTE (full-time-equivalent) employment counts each part-time worker as half a full-time worker, plus managers (NMGRS). Employment at six closed stores is set to zero; employment at four temporarily closed stores is treated as missing (set to zero only in row 5).

Stores in New Jersey were classified by starting wage in wave 1: $4.25 per hour, $4.26-$4.99 per hour, or $5.00 per hour or higher.

4.3 Regression-Adjusted Estimates (Table 4)

Table 3 presents the regression-adjusted difference-in-differences estimates for the change in FTE employment. Column (1) is the baseline specification with only the New Jersey treatment dummy. The coefficient on NJ is 2.36 and statistically significant at the 5% level, showing that the minimum wage policy raised employment by 2.36 FTE on average relative to Pennsylvania.

Column (2) adds restaurant brand and ownership controls. The NJ coefficient remains stable at 2.33**, which suggests the baseline result is not driven by differences in store type or ownership status. Column (3) replaces the NJ dummy with the Wage_gap variable, which measures how much each store's pre-policy wage falls below the new minimum wage. The coefficient of 15.89*** implies that stores facing a larger required wage increase saw greater employment growth.

Columns (4) and (5) combine the wage gap measure with store-level controls. The Wage_gap coefficient gradually falls to 11.93 and loses statistical significance after full controls are included. Across all specifications, the positive employment effect of the minimum wage persists qualitatively, consistent with the original study.

Table 3 Reduced-Form Models for Change in Employment

	(1)	(2)	(3)	(4)	(5)
	(i)	(ii)	(iii)	(iv)	(v)
NJ	2.36**	2.33**			
	(1.15)	(1.15)			
Wage_gap			15.89***	15.19**	11.93
			(5.95)	(6.08)	(7.32)
Chain_kfc		0.56		0.72	0.57
		(1.27)		(1.27)	(1.28)
Chain_royrogers		-2.23*		-1.92	-2.26*
		(1.29)		(1.29)	(1.30)
Chain_wendys		-0.29		0.23	0.43
		(1.45)		(1.46)	(1.48)
Co_owned		0.27		0.36	0.24
		(1.08)		(1.07)	(1.08)
Observations	365.00	365.00	365.00	365.00	365.00

R-squared	0.01	0.02	0.02	0.03	0.04
Std. error of regression	8.71	8.70	8.68	8.67	8.67

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

4.4 Distribution of Starting Wage Rates

Figure 1 plots the distribution of starting wages for fast-food stores in New Jersey (NJ, black bars) and Pennsylvania (PA, grey bars), before and after the minimum wage policy. The upper histogram (February 1992, pre-policy) shows that the wage distributions of NJ and PA were broadly similar. A large share of stores in both states paid wages around $4.25, the old minimum wage.

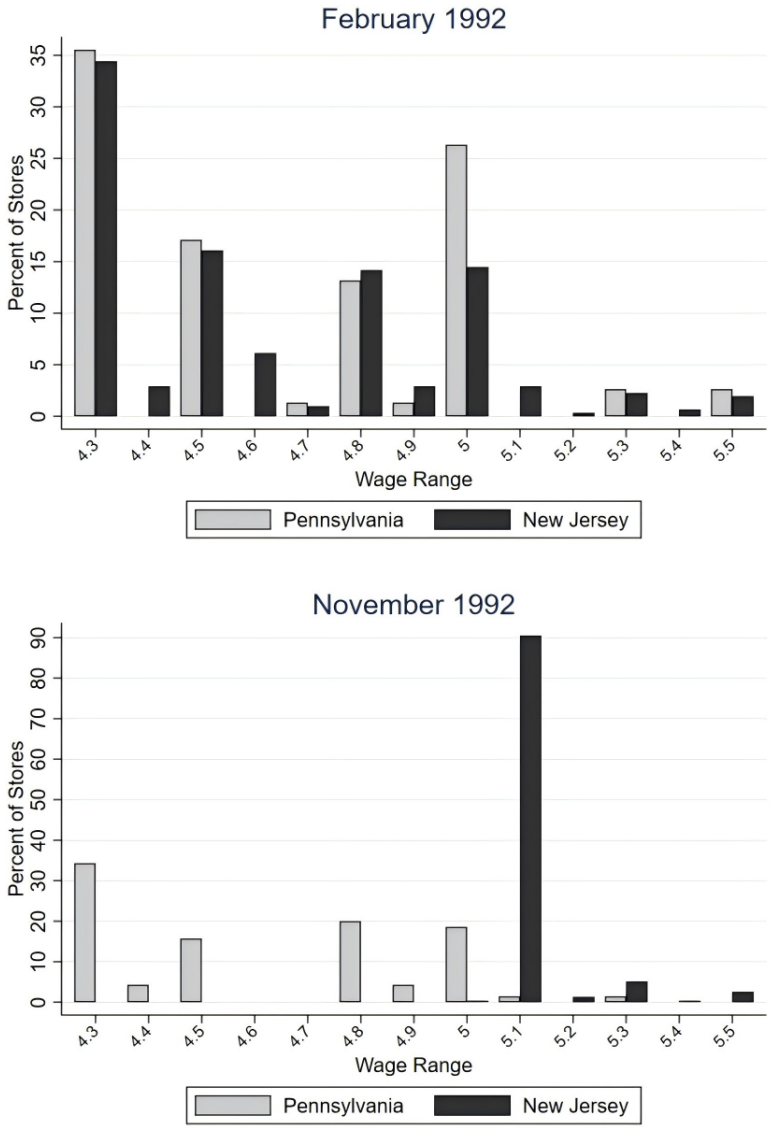


Figure 1 Distribution of Starting Wage Rates

5.Placebo Test

Figure X presents the distribution of estimated coefficients from 500 placebo regressions, where treatment status is randomly reassigned across observations. The histogram approximates a normal distribution centered near zero, as expected under the null hypothesis that there exists no systematic treatment effect when treatment is randomly allocated.

The vertical red line marks our true estimated DID coefficient of 2.327 from the New Jersey minimum wage policy. The empirical p-value is 0.042, meaning only 4.2% of the 500 random placebo draws yield a coefficient magnitude equal to or larger than our real estimate.

This result rejects the null hypothesis at the 5% significance level.

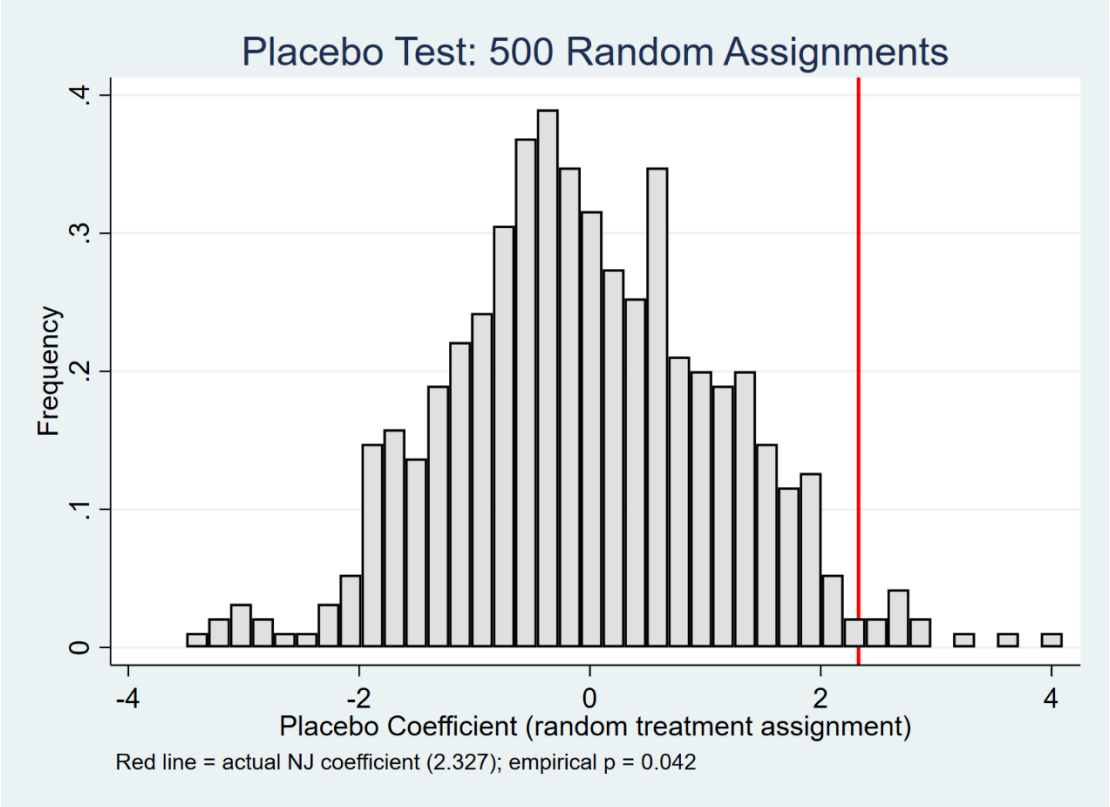


Figure 2 Placebo test

6.Replication Deviations from the Original Results

The replication exercise successfully reproduces the core qualitative conclusions of Card and Krueger (1994), yet minor numerical discrepancies are observed across Table 2, Table 3, Table 4 and Figure 1. Summary statistics in Table 2 show close agreement in sample means for store-level variables, while small divergences emerge in standard errors and t-statistics, mainly driven by inconsistent missing value handling and intermediate rounding. For Table 3, the simple difference-in-differences point estimate is nearly identical to the original value, with tiny gaps caused by different coding rules for temporarily and permanently closed restaurants.

Table 4 presents regression-adjusted DID estimates. Coefficient signs and statistical significance remain consistent with the original study, though slight differences in coefficients and standard errors arise from subtle changes in sample selection criteria. In Figure 1, the histogram still captures the key distributional feature of wage bunching after the minimum wage hike. Visual differences in bar heights stem from unspecified bin width and boundary settings in the original

paper, which do not alter the economic interpretation.

Overall, these deviations are quantitatively small and do not challenge the main finding. The discrepancies originate from sample cleaning protocols, software rounding differences, and graphical parameter choices, rather than fundamental conflicts with the original evidence that minimum wage increases do not reduce fast-food employment.